

PRODUCT WHITE PAPER

# Scrum Health Intelligence

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*From retrospective reporting to early, explainable delivery decisions*

**PRODUCT THESIS** An AI-enabled intelligence layer for Jira and Confluence that detects delivery risk, maps dependencies, automates leadership reporting, and helps teams act before sprint and release outcomes deteriorate.

**Audience**

Engineering and Product Leaders

**Product concept**

Jira + Confluence cloud integration

**Version**

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# Executive summary

Most delivery dashboards describe what has already happened. They aggregate counts, velocity, burndown, and status colors, but require leaders to reconstruct the causal story: Which commitment is at risk? What changed? Which dependency is blocking progress? How reliable is the signal? Who should act next?

Scrum Health Intelligence addresses this decision gap. It continuously interprets structured work signals from Jira and contextual evidence from Confluence, then converts them into an explainable view of sprint-goal confidence, dependency exposure, execution flow, readiness, and reporting burden. The product does not judge individual performance or replace Scrum events. It helps teams inspect and adapt with better evidence.

**NORTH-STAR OUTCOME** Increase the percentage of material delivery risks identified with enough lead time for a team to mitigate them before a sprint or release commitment is missed.

## The opportunity

| Current state                                         | Product intervention                                  | Measurable result                                     |
|-------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------|
| Status assembled manually across Jira and Confluence  | Generate role-specific narratives with evidence links | Lower reporting effort and shorter decision cycles    |
| Dependencies discovered in meetings or after slippage | Infer, validate, and monitor a dependency graph       | Earlier escalation and fewer dependency-driven misses |
| Risk is subjective or reduced to red/amber/green      | Combine transparent rules, trends, and AI evidence    | Higher signal precision and stakeholder trust         |
| Executives see activity, not outcome confidence       | Roll up product-goal and release exposure             | Clearer portfolio trade-offs and interventions        |

## Recommendation

Enter the market through a narrow wedge: automated sprint and release health briefs for Jira/Confluence customers. Earn trust with cited evidence and editable assessments; then expand into cross-team dependency intelligence, scenario analysis, and portfolio decision support.

# 1. Problem framing: dashboards without decisions

Engineering organizations rarely lack data. They lack a reliable translation from work-system exhaust into timely decisions. Jira records issue state and transitions; Confluence contains goals, decisions, plans, risks, and meeting context. The intelligence needed to assess delivery health is split across both systems and often reconciled manually.

## Four high-value problems

| Problem                         | Root cause                                                | Business consequence                                 |
|---------------------------------|-----------------------------------------------------------|------------------------------------------------------|
| Late risk detection             | Lagging metrics and status-by-opinion                     | Reduced mitigation window; surprise misses           |
| Invisible dependency exposure   | Links are incomplete; ownership and dates are implicit    | Blocked work, launch coordination failures           |
| Manual reporting                | Different audiences need different summaries              | Hours spent collecting, formatting, and explaining   |
| Weak executive product tracking | Activity metrics are disconnected from goals and releases | Slow trade-offs; false confidence or over-escalation |

## Jobs to be done

- When a sprint or release starts drifting, help a delivery leader understand the cause, confidence, impact, and next-best action before the commitment is missed.
- When multiple teams share a milestone, help a product leader see which dependency paths threaten the outcome and where escalation creates the most leverage.
- When leadership needs an update, produce a concise, evidence-linked narrative without forcing teams to recreate information in slides or spreadsheets.

## Non-goals

- Ranking people, monitoring individual activity, or using velocity as a productivity target.
- Replacing Product Owners, Scrum Masters, retrospectives, or human accountability.
- Presenting model-generated claims without source evidence, confidence, or an override path.

## 2. Product definition and positioning

Scrum Health Intelligence is a system of intelligence—not a second system of record. Jira remains authoritative for work state; Confluence remains authoritative for narrative context and decisions. The product observes changes, normalizes them into a delivery model, evaluates health, and publishes role-specific insights back into existing workflows.

**POSITIONING** For engineering and product organizations using Jira and Confluence, Scrum Health Intelligence is an explainable delivery-risk platform that turns fragmented execution data into early warnings and action-ready leadership reporting—without requiring teams to maintain another status system.

### Primary users and value

| Persona              | Critical question                          | Product value                                                 |
|----------------------|--------------------------------------------|---------------------------------------------------------------|
| VP Engineering / CTO | Which commitments need intervention now?   | Portfolio exposure, trend, confidence, and accountable action |
| Product leader       | Are product goals and releases converging? | Outcome-linked delivery confidence and dependency paths       |
| Engineering manager  | Why is flow or predictability degrading?   | Contributing signals, affected scope, and mitigation options  |
| Scrum Master / TPM   | What needs facilitation or escalation?     | Automated brief, blocker aging, dependency follow-through     |
| Team                 | What changed and what can we do next?      | Transparent, correctable signals—not surveillance             |

### Product principles

1. Decision-first: every surfaced signal must map to a decision, owner, or follow-up.
2. Evidence before eloquence: concise narratives must cite Jira fields, change history, or Confluence passages.
3. Team-safe: evaluate systems and commitments, never infer individual performance or intent.
4. Configurable semantics: respect each team's workflow, Definition of Done, and operating cadence.
5. Progressive automation: begin with recommendations; automate actions only after trust and controls mature.

## 3. The health model

A single opaque score creates false precision. The product should expose a small set of interpretable dimensions, a directional trend, confidence, and the evidence that changed the assessment. A composite health index can support portfolio sorting, but should never hide its components.

| Dimension           | Example signals                                                    | Decision enabled                 |
|---------------------|--------------------------------------------------------------------|----------------------------------|
| Goal confidence     | Sprint goal coverage; critical scope state; acceptance readiness   | Re-scope, swarm, or protect goal |
| Flow health         | Cycle-time shift; WIP age; blocked duration; carryover             | Remove bottleneck or reduce WIP  |
| Predictability      | Commitment volatility; completion forecast; historical calibration | Reset expectation or change plan |
| Dependency exposure | Unresolved predecessors; owner/date gaps; critical-path slack      | Escalate, sequence, or decouple  |
| Quality/readiness   | Reopen trend; defect burden; unresolved release criteria           | Harden, delay, or adjust launch  |
| Alignment hygiene   | Goal freshness; decision recency; Jira/Confluence contradiction    | Clarify goal or update source    |

### Transparent scoring

For dimension  $d$  at time  $t$ , normalize each signal to a 0–100 risk scale and compute:

$$\text{Risk}(d,t) = \sum w_i \times \text{signal}_i(t) \quad | \quad \text{Health} = 100 - \text{Risk}$$

Weights are organization-configurable and validated against historical outcomes. The product displays the top contributing signals, change since the prior observation, data freshness, and model confidence. Portfolio rollups weight outcome exposure—not team size or story-point volume.

### Dependency risk

Represent initiatives, epics, releases, and key work items as a directed graph. For dependency edge  $e$ :

$$\text{Exposure}(e) = P(\text{delay}) \times \text{Impact} \times \text{Criticality} \times (1 - \text{Mitigation coverage})$$

AI may propose missing edges from text and behavior, but a proposed dependency remains unconfirmed until an authorized user validates it. Criticality is derived from path slack and downstream reach; impact is tied to a product goal, release, or customer commitment.

## 4. Experience: insight to action

### Team health brief

A daily or on-demand brief shows what changed, why it matters, and what requires attention. It emphasizes the sprint goal, not raw completion. Each statement links to evidence and supports confirm, dismiss, annotate, or assign actions.

| Signal                                          | Evidence                                                          | Suggested response                                         |
|-------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------|
| Sprint-goal confidence fell from high to medium | Two goal-critical items blocked >48h; test environment date moved | Validate environment recovery; identify decoupling option  |
| Carryover risk increased                        | Three items exceed the team's calibrated cycle-time range         | Swarm on oldest goal-critical item; de-scope non-goal work |
| Dependency needs confirmation                   | Confluence launch plan names Identity API; no Jira link exists    | Confirm owner, required-by date, and dependency edge       |

### Executive product view

- Outcome map: product goals → releases → initiatives → teams, with health and confidence at each level.
- Exception queue: only material changes, ranked by exposure and intervention leverage.
- Dependency pathways: cross-team risks with owner, required-by date, slack, and mitigation status.
- Narrative brief: what changed, business impact, decisions required, and evidence links.
- Trend and calibration: whether previous warnings were accurate and whether mitigations worked.

### The closed-loop workflow

| Observe                                             | Interpret                                        | Decide                                          | Learn                                       |
|-----------------------------------------------------|--------------------------------------------------|-------------------------------------------------|---------------------------------------------|
| Ingest Jira changes and approved Confluence context | Rules + models score signals and explain drivers | User confirms, assigns, mitigates, or dismisses | Outcome and feedback recalibrate thresholds |

**KEY UX RULE** Never generate a red status without showing: the affected outcome, material evidence, confidence, change driver, and next responsible action.

## 5. AI and data architecture

The architecture combines deterministic analytics with probabilistic AI. Rules and statistical models handle measurable state; language models extract context and draft narratives. This separation improves reproducibility, cost control, and trust.

| Layer          | Responsibility                                                               | Design choice                                       |
|----------------|------------------------------------------------------------------------------|-----------------------------------------------------|
| Source         | Jira work items, links, changelog;<br>Confluence pages and decisions         | Read-minimized, permission-aware connectors         |
| Ingestion      | Webhooks plus scheduled reconciliation                                       | Idempotent events; freshness monitoring             |
| Semantic model | Goals, sprints, releases, work, dependencies, risks, decisions               | Canonical IDs and temporal history                  |
| Signal engine  | Flow, aging, scope change, readiness, graph exposure                         | Deterministic and statistically calibrated features |
| AI reasoning   | Context extraction, contradiction detection, dependency proposals, summaries | Retrieval-grounded; structured outputs; citations   |
| Experience     | Team brief, executive view, alerts, Confluence reports                       | Role-based policy and feedback controls             |

### AI capability map

| Use case               | AI role                                                       | Control                                         |
|------------------------|---------------------------------------------------------------|-------------------------------------------------|
| Dependency discovery   | Extract candidate relationships, owners, dates, and rationale | Human confirmation before graph activation      |
| Risk explanation       | Summarize top feature contributions in business language      | Source citations; no unsupported causal claim   |
| Status automation      | Generate audience-specific briefs from the same facts         | Editable draft; approval and publication policy |
| Mitigation support     | Suggest options based on pattern, constraints, and precedent  | Recommendation only; display assumptions        |
| Data-quality detection | Identify stale goals, contradictions, and missing ownership   | Route to source owner for correction            |

## Jira and Confluence integration

Jira Cloud REST APIs expose issues and related platform resources, while Confluence Cloud REST API v2 exposes content resources. The connector should use least-privilege OAuth scopes, preserve source permissions during retrieval, support pagination and rate limits, and reconcile webhook events with periodic snapshots. API behavior and deprecations must be monitored through Atlassian's developer changelogs.



## 6. Trust, governance, and responsible use

Adoption will fail if teams experience the system as opaque surveillance. Trust is therefore a product capability, not a compliance appendix.

| Risk                    | Product safeguard                                                 | Proof mechanism                               |
|-------------------------|-------------------------------------------------------------------|-----------------------------------------------|
| False or noisy alerts   | Threshold calibration, suppression windows, severity gating       | Precision, recall, dismissals, alert burden   |
| Hallucinated narrative  | Retrieval-grounding and claim-level citations                     | Unsupported-claim test suite                  |
| Individual surveillance | No individual health scores; minimum aggregation rules            | Policy tests and audit logs                   |
| Permission leakage      | Source ACL propagation; tenant isolation; field-level controls    | Access-control tests and security review      |
| Automation bias         | Confidence, counter-evidence, editable assessment, human approval | Override rate and decision review             |
| Metric gaming           | Balanced dimensions; no velocity targets; outcome calibration     | Metric-health review and qualitative research |

### Explainability contract

- Every assessment identifies the affected goal or commitment.
- Every risk narrative points to the exact data or approved context used.
- Every model output includes confidence, freshness, and known missing data.
- Every user can correct a relationship, annotate context, or dismiss a signal with a reason.
- Every material change is auditable: input version, model/rule version, output, and user action.

### Data boundary

The product should default to metadata and narrowly retrieved page sections, not indiscriminate content indexing. Organizations configure approved Confluence spaces, retention, regional processing, and model-provider policies. Sensitive fields are excluded or tokenized before model inference where feasible.

## 7. Product analytics and business value

The measurement system must distinguish product usage from customer value and model quality. The north star is early, actionable risk detection—not dashboard views.

| Level      | Metric                                                                    | Why it matters                                    |
|------------|---------------------------------------------------------------------------|---------------------------------------------------|
| North star | Material risks identified before the mitigation cutoff                    | Measures usable lead time, not prediction theater |
| Outcome    | Dependency-driven miss rate; unplanned carryover; release surprise rate   | Tests whether delivery outcomes improve           |
| Efficiency | Hours of reporting effort avoided; brief preparation cycle time           | Quantifies manual-effort reduction                |
| Decision   | Median time from risk detection to owned action                           | Measures organizational response                  |
| Model      | Alert precision/recall; calibration error; evidence coverage              | Measures signal reliability                       |
| Adoption   | Weekly active leader/team cohorts; brief approval rate; action conversion | Measures habit and workflow fit                   |
| Guardrail  | Alert burden; override rate; team trust score; permission incidents       | Prevents harmful optimization                     |

### Value model

A pilot business case should separate direct labor savings from avoided delivery impact:

$$\text{Annual value} = \text{reporting hours avoided} \times \text{loaded labor rate} + \text{mitigated misses} \times \text{expected impact} - \text{platform cost}$$

Avoided-miss value should be presented as a range with explicit assumptions. Do not claim causality from simple before/after comparisons. Use matched teams or staggered rollout where possible, and track whether surfaced risks were both accurate and acted upon.

### Pilot evaluation design

6. Establish an 8–12 week baseline for reporting effort, dependency incidents, carryover, and surprise misses.
7. Launch with 3–5 teams sharing a meaningful release dependency network.
8. Pre-register thresholds for signal precision, action lead time, and reporting-time reduction.

9. Run weekly qualitative reviews to identify missing context, harmful signals, and workflow friction.
10. Compare change against baseline and a similar non-pilot cohort; document confounders.

## 8. Market strategy and differentiation

Jira already provides workflows, dependencies, dashboards, and AI capabilities. A marketable product cannot win by recreating those surfaces. The differentiation is a cross-source, outcome-centered intelligence model that learns organizational delivery patterns and closes the loop from risk evidence to owned mitigation and calibrated result.

| Alternative               | Strength                                      | Gap addressed by this concept                                         |
|---------------------------|-----------------------------------------------|-----------------------------------------------------------------------|
| Native dashboards         | Immediate, familiar reporting                 | Mostly configured views; limited cross-source causal narrative        |
| BI / data warehouse       | Flexible historical analysis                  | High setup burden; weak workflow and intervention loop                |
| Manual program reviews    | Rich judgment and context                     | Slow, inconsistent, expensive, hard to calibrate                      |
| Generic AI summaries      | Fast narrative generation                     | May summarize noise; lacks delivery semantics and outcome calibration |
| Scrum Health Intelligence | Evidence-linked risk and dependency decisions | Must prove precision, trust, and incremental value                    |

### Beachhead and expansion

- Beachhead: 200–2,000-person software organizations using Jira Cloud and Confluence Cloud, with multiple teams coordinating releases and recurring executive reporting.
- Buyer: VP Engineering, Head of Product Operations, or transformation leader. Champions: TPMs, engineering managers, Scrum Masters, and product operations.
- Initial monetization: annual platform fee plus active-team tiers; enterprise governance and portfolio analytics as premium capabilities.
- Expansion: release governance, product-goal portfolio intelligence, scenario planning, and additional work/document systems.

### Defensible assets

- A temporal delivery knowledge graph connecting goals, work, decisions, dependencies, risks, and outcomes.
- Organization-specific calibration data linking early signals and interventions to actual results.
- Embedded feedback loops and trust controls that improve precision and make switching costly.



## 9. MVP roadmap

| Phase                              | Scope                                                                                         | Exit criteria                                                                   |
|------------------------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 0 — Discovery (4–6 weeks)          | Workflow mapping; data-quality audit; 20–30 user interviews; offline historical analysis      | Three high-value decisions validated; data coverage and baseline established    |
| 1 — Trustworthy brief (8–12 weeks) | Jira ingestion; approved Confluence retrieval; sprint/release brief; evidence links; feedback | ≥80% material-claim evidence coverage; useful precision; reporting time reduced |
| 2 — Dependency intelligence        | Validated graph; missing-link proposals; owner/date gaps; path exposure                       | Earlier dependency escalation; lower unresolved critical-edge rate              |
| 3 — Portfolio decisions            | Goal/release rollup; exception queue; action tracking; calibration analytics                  | Leaders use insights in operating review; response time improves                |
| 4 — Assisted automation            | Policy-based briefs, routing, and follow-up; scenario analysis                                | Automation meets trust, security, and error-budget gates                        |

### MVP scope discipline

The MVP should solve one repeatable workflow end to end: produce a trusted weekly sprint/release health brief in under five minutes, highlight no more than the most material risks, and convert each accepted risk into an owned action. Dependency inference can begin in shadow mode to collect labels before becoming user-facing.

### Top product risks and tests

| Assumption                             | Fastest test                                     | Kill / pivot signal                                   |
|----------------------------------------|--------------------------------------------------|-------------------------------------------------------|
| Leaders act on earlier warnings        | Concierge briefs in existing review meetings     | Briefs are read but do not change decisions           |
| Jira/Confluence contain enough context | Historical reconstruction of 10–15 misses        | Most causes exist only in inaccessible channels       |
| Teams will correct AI signals          | Clickable evidence + one-step feedback prototype | High distrust or low correction participation         |
| Customers will pay beyond native tools | Pricing interviews and paid design-partner pilot | Value perceived as a dashboard feature, not a product |

# 10. Decision and conclusion

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The strongest version of this product is not an automated Scrum scorecard. It is an early-warning and decision-support layer that respects Scrum's empirical foundation: make work and evidence more transparent, improve inspection, and shorten the path to adaptation.

The commercial opportunity rests on a specific wedge: remove the recurring cost and ambiguity of delivery health reporting while finding material risks earlier. If the product can demonstrate credible lead time, accurate dependency exposure, and materially lower reporting effort—without eroding team trust—it can expand from a reporting assistant into a system of intelligence for product delivery.

**GO-FORWARD DECISION** Fund a design-partner discovery and offline-validation phase before building a broad dashboard. Prove three things first: sufficient source evidence, actionable risk lead time, and willingness to pay for the closed-loop workflow.

## Discovery questions

- Which executive decisions repeatedly require manual reconciliation of Jira and Confluence?
- How early must a risk be identified for mitigation to remain practical?
- Which dependency failures create the largest business impact, and how were they first visible?
- What evidence would make teams trust, correct, and act on an AI-generated assessment?
- Which native Jira capabilities are already sufficient, and where does cross-source intelligence create incremental value?

## Appendix A — Signal dictionary

| Signal                 | Definition                                                                | Caution                                        |
|------------------------|---------------------------------------------------------------------------|------------------------------------------------|
| Blocked age            | Elapsed working time in a configured blocked state                        | Status hygiene varies by team                  |
| Cycle-time deviation   | Current item age relative to team/issue-type baseline                     | Avoid comparison across unlike work            |
| Scope volatility       | Goal-linked scope added, removed, or materially changed after start       | Change may be intentional learning             |
| Carryover propensity   | Calibrated likelihood that committed work remains incomplete              | Never convert to individual score              |
| Dependency slack       | Time between forecasted prerequisite completion and required-by date      | Forecast uncertainty must propagate            |
| Ownership completeness | Critical risks/dependencies with named accountable owner and date         | Names indicate accountability, not performance |
| Goal coverage          | Share of goal evidence mapped to deliverable work and acceptance criteria | Mapping quality affects usefulness             |
| Evidence freshness     | Age of source data supporting the assessment                              | Stale evidence lowers confidence               |

## Appendix B — Example executive brief

**OVERALL** Release Atlas remains **MEDIUM** confidence. The primary exposure is an unconfirmed Identity API dependency affecting two goal-critical epics. Confidence is 0.76; source freshness is 94%.

- Changed: required-by date moved five working days earlier in the launch plan; Jira dependency links were not updated.
- Impact: forecast slack is now two days; downstream validation cannot begin until the API contract is stable.
- Decision: confirm whether to de-scope delegated administration or fund a parallel contract-mock path by Thursday.
- Owner/action: Product Director to confirm scope; Platform EM to validate mock feasibility.
- Evidence: linked release plan passage, Jira epics, issue changelog, and dependency proposal.



# References

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1. [Schwaber, K., and Sutherland, J. The Scrum Guide \(November 2020\).](#)
2. [Atlassian. Jira Cloud Platform REST API v3.](#)
3. [Atlassian. Confluence Cloud REST API v2.](#)
4. [Atlassian. Jira features: workflows, dependencies, dashboards, and AI.](#)
5. [Atlassian. Jira Cloud Platform changelog.](#)

Note: Product metrics, scoring formulas, positioning, and roadmap in this paper are proposed design hypotheses. They require customer discovery, historical validation, security review, and controlled pilots before production use.